TRANSVALS IN SEMIDISTRIBUTIVE LATTICES

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Abstract. We define transvals, which are a generalization of intervals in semidistributive lattices. We show that transvals of the Tamari lattice $\mathbf{Tam}_n$ are in bijection with convex topologies on $\{1, ..., n\}$, and that transvals of the weak order on a Weyl group W are in bijection with homogeneous right coideal subalgebras of its quantized enveloping algebra $U_q(\mathfrak{g})$.

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1. Introduction

Citer [Han24] [CR15] [BGP+26] [CPP19] [HK12]

2. Notations and preliminaries

2.1. **Nuclear intervals (faces). L** be a finite lattice.

Let $x \leq y \in \mathbf{L}$. We write

$$\mathrm{pop}_x(y) := y \wedge \bigwedge \{z \in \mathbf{L} : x \leq z \lessdot y\}$$
$$\mathrm{pop}^y(x) := x \vee \bigvee \{z \in \mathbf{L} : x \lessdot z \leq y\}.$$

$$\mathrm{pop}^{\downarrow}(y) := y \wedge \bigwedge \{x \in \mathbf{L} : x \lessdot y\},$$
$$\mathrm{pop}^{\uparrow}(x) := x \vee \bigvee \{y \in \mathbf{L} : x \lessdot y\}.$$

When j is a join-irreducible element of $\mathbf{L}$, we also write $j_* := \mathrm{pop}^{\downarrow}(j)$ for the only element covered by j. If m is meet-irreducible, we write $m^* := \mathrm{pop}^{\uparrow}(m)$.

Let $[x, y]$ be an interval of $\mathbf{L}$.

- We say that $[x, y]$ is *nuclear* if $x = \mathrm{pop}_x(y)$.
- We say that $[x, y]$ is *conuclear* if $y = \mathrm{pop}^y(x)$.
- We say that $[x, y]$ is *binuclear* if it is both nuclear and conuclear.

We also apply the same terminology to finite lattices, by seeing a lattice as its interval $[\mathbf{0}, \mathbf{1}]$.

2.2. **Semidistributive lattices.** A lattice is *meet-semidistributive* if for all $x, y, z \in \mathbf{L}$, $x \wedge y = x \wedge z$ implies $x \wedge (y \vee z) = x \wedge y$. The following assertions are equivalent:

- $\mathbf{L}$ is meet-semidistributive.
- The map $\kappa : \mathcal{J}(\mathbf{L}) \to \mathcal{M}(\mathbf{L})$ sending j to $\kappa(j) := \max\{z : z \wedge j = j_*\}$ is well-defined.
- Each element of $\mathbf{L}$ has a unique canonical meet-representation.

A lattice respecting the dual condition is called *join-semidistributive*. A lattice which is both meet- and join-distributive is called *semidistributive*.

2.3. **Join-distributive lattices.** The following assertions are equivalent:

- $\mathbf{L}$ is join-distributive.
- $\mathbf{L}$ is meet-semidistributive and upper-semimodular.
- For all $j \in \mathcal{J}(\mathbf{L})$, there is a unique $m \in \mathcal{M}(\mathbf{L})$ such that $j \wedge m = j_*$.

Any finite lattice $\mathbf{L}$ can be represented as the lattice of lower sets of a set-colored poset [HN18], which is the poset $\mathcal{J}(\mathbf{L})$ with coloring $\gamma(j) = \{m \in \mathcal{M}(\mathbf{L}) : j \wedge m = j_*\}$. The last assertion means that join-distributive lattices are the lattices such that $|\gamma(j)| = 1$, or equivalently $\gamma(j) = \{\kappa(j)\}$.

3. Definition of transvals and first results

3.1. **Faces.**

Lemma 3.1. *A finite meet-semidistributive lattice is a nuclear interval if and only if for all atoms w of $\mathbf{L}$, $\kappa(w)$ is a coatom of $\mathbf{L}$.*

Proof. It is straightforward to check that for all $x \gtrdot \mathbf{0}$, $x = \bigwedge_{y \gtrdot \mathbf{0}, y \neq x} \kappa(y)$ and $\mathbf{0} = \bigwedge_{y \gtrdot \mathbf{0}} \kappa(y)$ (these are the canonical meet-representations of x and $\mathbf{0}$, respectively). Hence, if for all atoms x , $\kappa(x)$ is a coatom, then $\mathbf{L}$ is a nuclear interval. Conversely, if there exists $x \gtrdot \mathbf{0}$ such that $\kappa(x)$ is not a coatom, it implies that for all coatoms $y \lessdot \mathbf{1}$, we have $x \leq y$, so their meet is above x and $\mathbf{L}$ is not nuclear. □

Proposition 3.2. *Let $\mathbf{L}$ be a finite meet-semidistributive lattice. Let $[x, y]$ be a conuclear interval of $\mathbf{L}$. Then $[x, y]$ is a nuclear interval.*

Proof. Without loss of generality, we take $\mathbf{L} = [x, y]$. Since $\mathbf{L}$ is conuclear, for all atoms $z \gtrdot x$, there exists $w \lessdot y$ such that $z \not\leq w$ (if not, the join of the atoms would be less than w), and then $z \wedge w = x$ so $w = \kappa(z)$. By Lemma 3.1, $[x, y]$ is nuclear. □

The converse is not true (see Figure 1). It follows from Proposition 3.1, and from its dual counterpart for join-semidistributive lattices, that in a semidistributive lattice, nuclear and conuclear intervals are the same. We will call *faces* the binuclear intervals of a semidistributive lattice, as they coincide with geometric faces of many polytopal realizations of lattices such as the weak order and the Tamari lattice. However, faces of a semidistributive lattice do not always have a geometric interpretation (see Figure 1).

We have seen in Lemma 3.1 that in a nuclear interval $[x, y]$, we have an injection from the atoms to the coatoms of $[x, y]$. In semidistributive lattices this is a bijection. We call the *dimension* of a face $[x, y]$ its number of atoms of this interval, or equivalently its number of coatoms.

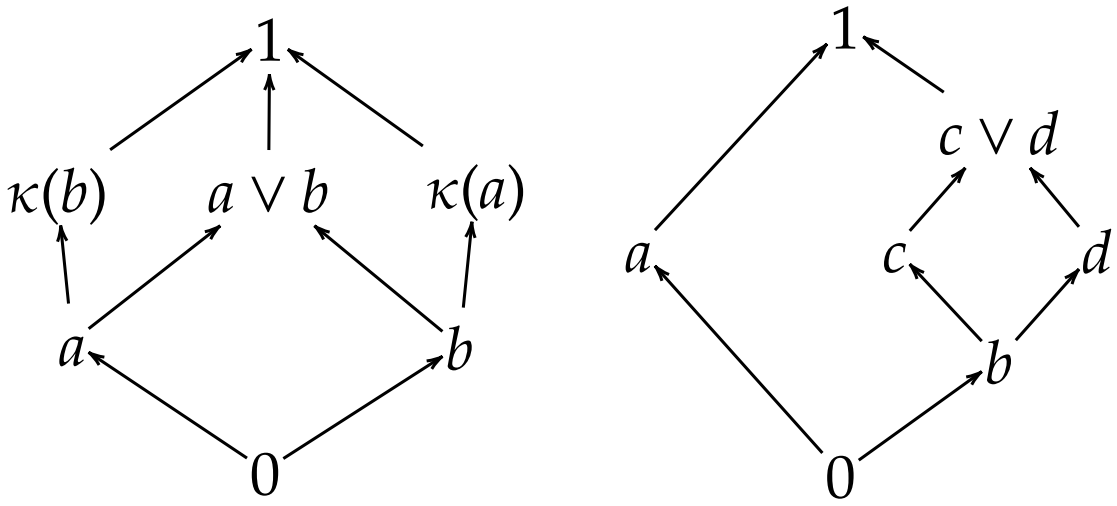


FIGURE 1. Left: a meet-semidistributive lattice which is nuclear but not conuclear. Right: A semidistributive lattice with two faces of dimension 2, one containing the other.

The join of two nuclear intervals is given by $[x_1, y_1] \vee [x_2, y_2] = \left[\mathrm{pop}^{\downarrow}_{x_1 \vee x_2}(y_1 \vee y_2), y_1 \vee y_2\right]$. Hanson proved that for any lattice, if its nuclear intervals compared coordinatewise form a lattice then their join has this expression [Han24].

Lemma 3.3. *Suppose* $\mathbf{L}$ *is finite and semidistributive. Let* $a_1, a_2, b_1, b_2 \in \mathbf{L}$.

- *If* $[a_1, b_1]$ *and* $[a_1, b_2]$ *are faces, then* $[a_1, b_1 \vee b_2]$ *is a face.*
- *If* $[a_1, b_1]$ *and* $[a_2, b_1]$ *are faces, then* $[a_1 \wedge a_2, b_1]$ *is a face.*

Proof. It suffices to prove only the first statement. Let $[a_1, b_1], [a_1, b_2]$ be faces. That is, $\mathrm{pop}^{\downarrow}_{a_1}(b_1) = a_1$ and $\mathrm{pop}^{\downarrow}_{a_1}(b_2) = a_1$. It is clear that $[a_1, b_1 \vee b_2]$ is an interval. By Proposition 3.2 and using that $\mathbf{L}$ is finite and join-semidistributive, it suffices to show $[a_1, b_1 \vee b_2]$ is a nuclear interval. That is, we want to show $\mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2) = a_1$.

Suppose $\mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2) \neq a_1$ for sake of contradiction. By definition, $\mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2) \geq a_1$. By assumption, we have $\mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2) > a_1$. Take any $w \in [a_1, b_1 \vee b_2]$ such that

$$a_1 \lessdot w \leq \mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2).$$

Let $i \in \{1, 2\}$. Consider the set

$$S_i := \{b_i\} \cup \{z \in \mathbf{L} : a_1 \leq z \lessdot b_i\}$$

so that $\mathrm{pop}^{\downarrow}_{a_1}(b_i) = \bigwedge S_i$. Now, either $w \leq b_i$ or $w \not\leq b_i$.

- Suppose $w \leq b_i$. Suppose $c_i \geq w$ for all $c_i \in S_i$. Then,

$$w > a_1 = \mathrm{pop}^{\downarrow}_{a_1}(b_i) = \bigwedge S_i \geq w.$$

  A contradiction. So we may choose $c_i \in S_i$ such that $c_i \not\geq w$. Moreover, by the assumption $w \leq b_i$ and using that $c_i \not\geq w$, we have

$$c_i < c_i \vee w \leq b_i.$$

  Since $c_i \lessdot b_i$, we have $c_i \vee w = b_i$.
- Suppose $w \not\leq b_i$. Pick $c_i = b_i$ so that $c_i \not\geq w$. By the choice of c_i, we have $c_i \vee w \geq b_i$.

In either case, there exists $c_i \in S_i$ with the properties $c_i \not\geq w$ and $c_i \vee w \geq b_i$ for both $i \in \{1, 2\}$. Using the first property $c_i \not\geq w$, we have

$$a_1 \leq c_i \wedge w < w.$$

Since $a_1 \lessdot w$, we have $c_i \wedge w = a_1$. By meet-semidistributivity, we have $(c_1 \vee c_2) \wedge w = a_1$. Since $w \leq b_1 \vee b_2$, we have $(b_1 \vee b_2) \wedge w = w \neq a_1$. Hence, $c_1 \vee c_2 \neq b_2 \vee b_2$. In particular,

$$a_1 \leq c_1 \vee c_2 < b_1 \vee b_2.$$

So there exists $c \in [a_1, b_1 \vee b_2]$ such that $c_1 \vee c_2 \leq c \lessdot b_1 \vee b_2$. By definition of w, we have $w \leq \mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2) \leq c$. Hence,

$$(c_1 \vee c_2) \vee w \leq c \lessdot b_1 \vee b_2.$$

Now, using the second property $c_i \vee w \geq b_i$, we have $(c_1 \vee c_2) \vee w \geq b_1 \vee b_2$. A contradiction. □

The proof of Lemma 3.3 can probably be shorter. The main idea is that $\mathrm{pop}^{\downarrow}_{a_1}(b_1 \vee b_2) = a_1$ if $\mathrm{pop}^{\downarrow}_{a_1}(b_1) = \mathrm{pop}^{\downarrow}_{a_1}(b_2) = a_1$. This follows from the simpler statement that $\mathrm{pop}^{\downarrow}(b_1 \vee b_2) = 0$ if $\mathrm{pop}^{\downarrow}(b_1) = \mathrm{pop}^{\downarrow}(b_2) = 0$. This only needs meet-semidistributivity. Join-semidistributive and finiteness is needed to use Proposition 3.2 We should check if join-semidistributivity is actually needed.

— J.

3.2. **Transvals.**

Definition 3.4. *Let* $\mathbf{L}$ *be a lattice, and* $x, y \in \mathbf{L}$. *The pair* (x, y) *is a transval of* $\mathbf{L}$ *if* $x_* \leq y$ *and* $x \leq y^*$.

Definition 3.5. *Let* $\mathbf{L}$ *be a lattice, and* $x, y \in \mathbf{L}$. *The pair* (x, y) *is an exact transval of* $\mathbf{L}$ *if* $[x \wedge y, x]$ *is conuclear and* $[y, x \vee y]$ *is nuclear.*

It is clear from the definition that exact transvals are transvals, however the converse is not true in general.

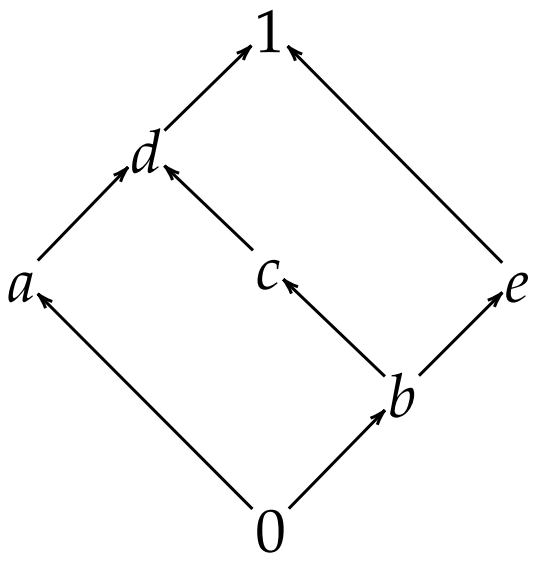


FIGURE 2. A lattice in which $(a, b), (d, b)$, and (d, e) are non-exact transvals.

In the following, $\mathbf{L}$ is a finite semidistributive lattice.

Definition 3.6. *$K_{\mathbf{L}}$ is the set of atoms $a \in \mathbf{L}$ such that $\kappa(a)$ is a coatom of $\mathbf{L}$.*

Proposition 3.7. *Let $\mathbf{L}$ be a finite semidistributive lattice. Transvals of $\mathbf{L}$ are the pairs $\left(\bigvee J, \bigwedge \kappa_{[x,y]}(J)\right)$ where $[x, y]$ is an interval of $\mathbf{L}$ and $J \subseteq K_{[x,y]}$.*

Proof. It is straightforward to show that $\mathrm{pop}^{\downarrow}\left(\bigvee J\right) \leq x \leq y$ and $\mathrm{pop}^{\uparrow}\left(\bigwedge \kappa(J)\right) \geq y \geq x$. Conversely, if (a, b) is a transval of $\mathbf{L}$ with $x := a \wedge b$ and $y := a \vee b$, for all c such that $x \lessdot c \leq a$, there exists $d \lessdot y$ such that $c \not\leq d$ (if not, we would have $c \leq \bigwedge_{b \leq d \lessdot y} d = b$ and $c \leq x$). Hence we have $b \leq \kappa_{[x,y]}(c) \lessdot y$. Dually, each coatom of $[x, y]$ which is above b corresponds to an atom above a, implying that $(a, b) = (\bigvee J, \bigvee \kappa[x, y](J))$ where $J = \{c : x \lessdot c \leq a\}$ and we have $\kappa[x, y](J) = \{d : b \leq d \lessdot y\}$, so $J \subseteq K_{[x,y]}$. □

Corollary 3.8. *All transvals of a finite semidistributive lattice are exact transvals.*

Corollary 3.8 states that each transval (x, y) of a semidistributive lattice corresponds to an interval in the lattice of faces $[x \wedge y, x] \leq [y, x \vee y]$. Conversely, for all faces $[x_1, x_2] \leq [y_1, y_2]$, the pair (x_2, y_1) is a transval.

Corollary 3.9. *Transvals of a finite semidistributive $\mathbf{L}$ are in bijection with triples (x, y, J) where $x \leq y \in \mathbf{L}$ and $J \subseteq K_{[x,y]}$.*

Corollary 3.9 is useful for enumeration purposes, as $|\mathbf{Trv}(\mathbf{L})| = \sum_{x \leq y} 2^{|K_{[x,y]}|}$. Other methods for enumerating transvals will be discussed in Section 6.

Proposition 3.10. *Let $[x, y]$ be an interval of $\mathbf{L}$. (y, x) is a transval of $\mathbf{L}$ if and only if $[x, y]$ is a nuclear interval.*

In a join-semidistributive lattice, there is a canonical labelling of edges by join-irreducible elements $\gamma : (x, y) \mapsto \min\{j : j \vee x = y\}$. $\Gamma(y) := \{\gamma(x, y) : x \lessdot y\}$ is the canonical join-representation of y, and we write $U(x) := \{\gamma(x, y) : y \gtrdot x\}$. Note that $\kappa(U(x))$ is the canonical meet-representation of x.

Also true for faces, as well as Lemma 3.13

— L.

Lemma 3.11. *Let $a_1, a_2, b_1, b_2 \in \mathbf{L}$.*

- *If (a_1, b_1) and (a_1, b_2) are transvals, then $(a_1, b_1 \vee b_2)$ is a transval.*
- *If (a_1, b_1) and (a_2, b_1) are transvals, then $(a_1 \wedge a_2, b_1)$ is a transval.*

This proposition could be proved more directly using Proposition 3.7 I think
— L.

Proposition 3.12. *(x, y) is a transval of $\mathbf{L}$ if and only if $\Gamma(x) \subseteq \{j \in \mathcal{J}(\mathbf{L}) : j \leq y\} \cup U(y)$*

Proof. If (x, y) is a transval, each $j \in \Gamma(x)$ satisfies $j_* \leq \mathrm{pop}^{\downarrow}(x) \leq y$ and $j \leq x \leq \mathrm{pop}^{\uparrow}(y)$. This tells us that either $j \leq y$ or $j \in U(y)$. Conversely, suppose that $\Gamma(x) \subseteq \{j \in \mathcal{J}(\mathbf{L}) : j \leq y\} \cup U(y)$. Since for all $j \in \Gamma(x)$, $j \leq y$ or $j \leq z$ for some $z \twoheadrightarrow y$, we have $x \leq \mathrm{pop}^{\uparrow} y$. Suppose, for a contradiction, that $\mathrm{pop}^{\downarrow}(x) \not\leq y$. Let $k \leq \mathrm{pop}^{\downarrow}(x)$ that is not below y, minimal with this property. Then k is join-irreducible and $k_* \leq y$. This gives us $k \in U(y)$, but $k \notin \Gamma(x)$, because $k \leq \mathrm{pop}^{\downarrow}(x)$. Let $z = y \vee k$, an upper cover of y. For every $j \in \Gamma(x)$, either $j \leq y$, or j labels a different upper cover of y. In either case, $z \wedge (y \vee j) = y$. Applying meet-semidistributivity over the finite set $\Gamma(x)$ gives $z \wedge (y \vee x) = y$. However, $k \leq \mathrm{pop}^{\downarrow}(x) \leq x$, so $z = y \vee k \leq y \vee x$. The left side is therefore $z > y$, a contradiction. Thus $\mathrm{pop}^{\downarrow}(x) \leq y$, proving the result. □

Lemma 3.13. *Let $a_1, a_2, b_1, b_2 \in \mathbf{L}$.*

- *If (a_1, b_1) and (a_2, b_1) are transvals, then $(a_1 \vee a_2, b_1)$ is a transval.*
- *If (a_1, b_1) and (a_1, b_2) are transvals, then $(a_1, b_1 \wedge b_2)$ is a transval.*

Proof. We use the following facts:

(1) For all $j \in \Gamma(x)$, $j_* \leq \mathrm{pop}^{\downarrow}(x)$, but $j \not\leq \mathrm{pop}^{\downarrow}(x)$;
(2) If j is join-irreducible and $j_* \leq y, j \leq \mathrm{pop}^{\uparrow}(y)$, and $j \not\leq y$, then $j \in U(y)$.

Fact (1) is proved as follows. If j labels $p \prec x$, then $j_* \leq p$. Every other lower cover q of x satisfies $p \vee q = x$, so $j \leq q$. Taking the meet of the lower covers proves the first assertion; $j \not\leq p$ proves the second.

Fact (2) is proved as follows. If no upper cover z of y contained j, we would have $j \wedge z = j_*$ for every such cover. Meet-semidistributivity would then give $j \wedge \mathrm{pop}^{\uparrow}(y) = j_*$, contradicting $j \leq \mathrm{pop}^{\uparrow}(y)$. Thus some upper cover contains j, and $j_* \leq y$ ensures that j is its label.

Finally, the allowed set of join-irreducibles on the right of Proposition 3.12 is downward closed among join-irreducibles: if $j \in U(y)$, every element strictly below j lies below $j_* \leq y$.

If (a, y) and (c, y) are transvals, the refinement property says that every label in $\Gamma(a \vee c)$ lies below some label in $\Gamma(a) \cup \Gamma(c)$. It therefore remains in that allowed set. By Proposition 3.12, $(a \vee c, y)$ is a transval.

Applying the same argument to the dual lattice proves the statement for the meets, completing the proof. □

Proposition 3.14. *Let (a_1, b_1) and (a_2, b_2) be transvals in $\mathbf{L}$. Let*

$$c = \min \{x \geq b_1 \vee b_2 : (a_1 \vee a_2, x) \text{ is a transval}\},$$
$$d = \max \{x \leq a_1 \wedge a_2 : (x, b_1 \wedge b_2) \text{ is a transval}\}.$$

Then,

$$(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee a_2, c),$$
$$(a_1, b_1) \wedge (a_2, b_2) = (d, b_1 \wedge b_2).$$

Proof. It suffices to prove $(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee a_2, c)$ for c as defined above. By Lemma 3.13 and using that $\mathbf{L}$ is finite, we have $(a_1 \vee a_2, c)$ is a transval.

First, we show $(a_1 \vee a_2, c)$ is an upper bound. That is, we want to show $(a_1 \vee a_2, c) \geq (a_i, b_i)$ for all $i \in \{1, 2\}$. By definition of c, we have that $c \geq b_1 \vee b_2$. So, $c \geq b_i$. Also, $a_1 \vee a_2 \geq a_i$. Hence, $(a_1 \vee a_2, c) \geq (a_i, b_i)$.

It remains to show $(a_1 \vee a_2, c)$ is a *minimal* upper bound. Take any transval (e, f) such that $(e, f) \geq (a_i, b_i)$ for all $i \in \{1, 2\}$. We want to show $(a_1 \vee a_2, c) \leq (e, f)$. By definition of transval,

$$a_{i*} \leq b_i \leq f \text{ and } a_i \leq e \leq f^*.$$

Hence, (a_i, f) is a transval for all $i \in \{1, 2\}$. By Lemma 3.13, $(a_1 \vee a_2, f)$ is a transval. By assumption, $f \geq b_1 \vee b_2$. So by definition of c, we have $c \leq f$. Also by assumption, $a_1 \vee a_2 \leq e$. Hence, $(a_1 \vee a_2, c) \leq (e, f)$. □

3.3. **Face transvals.**

Definition 3.15. *A pair $(x, y) \in \mathbf{L}^2$ is a face transval of $\mathbf{L}$ if it is a transval such that $[x \wedge y, x \vee y]$ is a face.*

4. Lattice structures of transvals

Should face transvals be named transfaces ? If so, should faces become interfaces ?? Meh.
— L.

Proposition 4.1. $\mathbf{FInt_L}$ *is a sublattice of* $\mathbf{FTrv_L}$*, and* $\mathbf{Int_L}$ *is a sublattice of* $\mathbf{Trv_L}$*.*

However, $\mathbf{FInt_L}$ is not a sublattice of $\mathbf{Int_L}$, and $\mathbf{FTrv_L}$ is not a sublattice of $\mathbf{Trv_L}$. $\mathbf{FTrv_L}$ contains another sublattice isomorphic to $\mathbf{FInt_L}$, consisting of pairs (y, x) such that $[x, y] \in \mathbf{FInt_L}$.

Proposition 4.2. *Join-irreducible elements of* $\mathbf{Int_L}$*,* $\mathbf{Trv_L}$*,* $\mathbf{FInt_L}$*, and* $\mathbf{FTrv_L}$ *are respectively the elements of the form:*

- $(0, j)$ *and* (j, j)*;*
- $(0, j)$ *and* (j, j_*)*;*
- (j_*, j) *and* (j, j)*;*
- (j_*, j) *and* (j, j_*)*;*

where 0 *is the minimal element of* $\mathbf{L}$*, and* $j \in \mathcal{J}(\mathbf{L})$*.*

Meet-irreducible elements of $\mathbf{Int_L}$*,* $\mathbf{Trv_L}$*,* $\mathbf{FInt_L}$*, and* $\mathbf{FTrv_L}$ *are respectively:*

- $(m, 1)$ *and* (m, m)*;*
- $(m, 1)$ *and* (m^*, m)*;*
- (m, m^*) *and* (m, m)*;*
- (m, m^*) *and* (m^*, m)*;*

where 1 *is the maximal element of* $\mathbf{L}$*, and* $m \in \mathcal{M}(\mathbf{L})$*.*

Proposition 4.3. *The* κ *maps of* $\mathbf{Int_L}$*,* $\mathbf{Trv_L}$*,* $\mathbf{FInt_L}$*, and* $\mathbf{FTrv_L}$ *are respectively given by:*

- $(0, j) \mapsto (m, m)$ *and* $(j, j) \mapsto (m, 1)$*;*
- $(0, j) \mapsto (m^*, m)$ *and* $(j, j_*) \mapsto (m, 1)$*;*
- $(j_*, j \mapsto (m, m)$ *and* $(j, j) \mapsto (m, m^*)$*;*
- $(j_*, j) \mapsto (m^*, m)$ *and* $(j, j_*) \mapsto (m, m^*)$*;*

where $m = \kappa(j)$*.*

4.1. **Cover relations.** Cover relations of $\mathbf{FInt}_{\mathbf{L}}$ are also of the form $(x, y_1) \lessdot (x, y_2)$ or $(x_1, y) \lessdot (x_2, y)$.

Cover relations of $\mathbf{Trv}_{\mathbf{L}}$ are also of the form $(x, y_1) \lessdot (x, y_2)$ or $(x_1, y) \lessdot (x_2, y)$. This comes from the fact that for all $(a, b) \leq (c, d) \in \mathbf{Trv}_{\mathbf{L}}$, we have $a_* \leq b \leq d$ and $a \leq c \leq d^*$, hence (a, d) is a transval of $\mathbf{L}$ with $(a, b) \leq (a, d) \leq (c, d)$.

However, cover relations of $\mathbf{FTrv}_{\mathbf{L}}$ are not always of this form (there is a counterexample for $\mathbf{Tam}_3$).

4.2. **Irreducibles graph.** Let $(\text{Ш}, \to)$ be the two-acyclic factorization system such that $\mathbf{L} \cong \mathrm{Pairs}(\to)$.

Definition 4.4. $\mathrm{Int}(\text{Ш})$, $\mathrm{FInt}(\text{Ш})$, $\mathrm{Trv}(\text{Ш})$ *and* $\mathrm{FTrv}(\text{Ш})$ *are the sets of pairs* (x, k) *with* $x \in \text{Ш}$ *and* $k \in \{0, 1\}$*, endowed with respective binary relations*

- $(x, 0) \to_{\mathrm{Int}} (x, 1)$ *for all* $x \in \text{Ш}$*, and* $x \to y$ *implies* $(x, k) \to_{\mathrm{Int}} (y, k)$ *for* $k \in \{0, 1\}$ *and* $(x, 0) \to_{\mathrm{Int}} (y, 1)$*;*
- $x \to y$ *implies* $(x, k) \to_{\mathrm{Trv}} (y, k)$ *for* $k \in \{0, 1\}$ *and* $(x, 0) \to_{\mathrm{Trv}} (y, 1)$*;*
- $(x, 0) \to_{\mathrm{FInt}} (x, 1)$ *for all* $x \in \text{Ш}$*, and* $x \to y$ *implies* $(x, k) \to_{\mathrm{FInt}} (y, k)$ *for* $k \in \{0, 1\}$*,* $(x, 0) \to_{\mathrm{FInt}} (y, 1)$*, and* $(x, 1) \to_{\mathrm{FInt}} (y, 0)$*;*
- $x \to y$ *implies* $(x, k) \to_{\mathrm{FTrv}} (y, k)$ *for* $k \in \{0, 1\}$*,* $(x, 0) \to_{\mathrm{FTrv}} (y, 1)$*, and* $(x, 1) \to_{\mathrm{FTrv}} (y, 0)$*.*

Proposition 4.5. *We have the following isomorphisms:*

- $\mathbf{Int}_{\mathbf{L}} \cong \mathrm{Pairs}(\to_{\mathrm{Int}})$.
- $\mathbf{FInt}_{\mathbf{L}} \cong \mathrm{Pairs}(\to_{\mathrm{FInt}})$.
- $\mathbf{Trv}_{\mathbf{L}} \cong \mathrm{Pairs}(\to_{\mathrm{Trv}})$.
- $\mathbf{FTrv}_{\mathbf{L}} \cong \mathrm{Pairs}(\to_{\mathrm{FTrv}})$.

4.3. **Spine of the lattice transvals in extremal lattices.** If $\mathbf{L}$ is trim, which is the case if and only if $\mathbf{L}$ is extremal [TW19, Thm. 1.4], the set of elements appearing in a chain of maximal length form a sublattice $\mathbf{Spine}(\mathbf{L})$ called the *spine* of $\mathbf{L}$, which is a distributive lattice [Tho06]. $\mathbf{L}$ is extremal if and only the corresponding factorization system is acyclic, and the induced order on $\mathcal{J}(\mathbf{L})$ is the poset of irreducibles of $\mathbf{Spine}(\mathbf{L})$:

Find citation ? I use the same result in my paper on Jurassian lattices.
— L.

Proposition 4.6. *Let* $\mathbf{L}$ *be a finite semidistributive extremal lattice, and let* $\mathcal{G}$ *be the poset obtained as the transitive closure of* $\to_{\mathbf{L}}$*. We have* $\mathbf{Spine}(\mathbf{L}) \cong \mathbf{Low}(\mathcal{G})$*.*

Proof. Elements of $\mathbf{Spine}(\mathbf{L})$ correspond to pairs $(X, Y) \in \mathrm{Pairs}(\to_{\mathbf{L}})$ such that $X \cup Y = \mathcal{J}(\mathbf{L})$, since chains of maximal length in $\mathbf{L}$ have length $|\mathcal{J}(\mathbf{L})|$ and (X, Y) belongs to chains of length at most $|X| + |Y|$. For all $(X, Y) \in \mathrm{Pairs}(\to_{\mathbf{L}})$, we have $X \cup Y = \mathcal{J}(\mathbf{L})$ if and only if $x \to_{\mathbf{L}} y$ and $x \in X$ imply $y \in X$, which is the case if and only X is a lower set of $\mathcal{G}$. □

Proposition 4.7. *Let* $\mathbf{L}$ *be a finite semidistributive extremal lattice. We have*

- $\mathbf{Spine}(\mathbf{Int}_{\mathbf{L}}) = \mathbf{Int}_{\mathbf{Spine}(\mathbf{L})}$*;*
- $\mathbf{Spine}(\mathbf{Trv}_{\mathbf{L}}) = \mathbf{Trv}_{\mathbf{Spine}(\mathbf{L})}$*;*
- $\mathbf{Spine}(\mathbf{FInt}_{\mathbf{L}}) = \mathbf{FInt}_{\mathbf{Spine}(\mathbf{L})}$*;*
- $\mathbf{Spine}(\mathbf{FTrv}_{\mathbf{L}}) = \mathbf{FTrv}_{\mathbf{Spine}(\mathbf{L})}$*.*

Proof. Direct proof or corollary of the description of TAFS. □

5. Spine intervals and spine transvals

Definition 5.1. *A spine interval of a trim lattice* $\mathbf{L}$ *is an interval* $[x, y]$ *intersecting its spine,* i.e.*, there exists* $x \leq z \leq y$ *such that* $z \in \mathbf{Spine}(\mathbf{L})$.

Spine intervals of a semidistributive and extremal lattice Pairs($\to$) form (conjecturally) a lattice Pairs($\to_{\mathrm{SpInt}}$), where SpInt(Ш) is the set of pairs (x, k) with $x \in$ Ш and $k \in \{0, 1\}$, endowed with the binary relation where $(x, 0) \to_{\mathrm{SpInt}} (y, 0)$ and $(x, 1) \to_{\mathrm{SpInt}} (y, 1)$ if $x \to y$, and $(x, 0) \to_{\mathrm{SpInt}} (y, 1)$ if $x \leq y$ in $\mathcal{G}$, the transitive closure of $\to$. Spine intervals of Tamari seem to coincide with infinitely modern intervals, which are counted by the same sequence as ternary trees [Rog18].

The faces intersecting the spine of such lattices does not always form a semidistributive lattice (consider for example $\mathbf{Camb}_{s_2s_1s_3}$, $\mathbf{Jura}_{132}$, or the Hochschild lattice $\mathbf{Tr}_3$). However the spine faces of the Tamari lattice form a semidistributive and extremal lattice, which is isomorphic to the lattice of nested topologies.

We define SpTrv(Ш) as the set of pairs (x, k) with $x \in$ Ш and $k \in \{0, 1\}$, endowed with the binary relation where $(x, 0) \to_{\mathrm{SpTrv}} (y, 0)$ and $(x, 1) \to_{\mathrm{SpTrv}} (y, 1)$ if $x \to y$, and $(x, 0) \to_{\mathrm{SpTrv}} (y, 1)$ if $x < y$ in $\mathcal{G}$.

n	1	2	3	4	5	6	7	8	9	10	11
$A007052$	1	3	10	34	116	396	1352	4616	15760	53808	183712
$A001764$	1	3	12	55	273	1428	7752	43263	246675	1430715	8414640
$A001353$	1	4	15	56	209	780	2911	10864	40545	151316	564719
$A121545$	1	4	17	81	412	2192	12049	67891	390041	2276176	13455356

Table 1. The number of spine faces, spine intervals, spine, spine face transvals and spine transvals of $\mathbf{Tam}_n$.

6. Canonical join-representations

Definition 6.1. *Let* G *be a directed graph. A min-max set of* G *is a subset* $X \subseteq G$ *such that for all* $x \in X$*, there is no* $y, z \in X$ *such that* $y \to x \to z$.

Definition 6.2. *An independant set of* G *is a subset* $X \subseteq G$ *such that* $x \in X$*, there is no* $y \in X$ *such that* $x \to y$ *or* $y \to x$.

Let $\mathbf{L}$ be a finite semidistributive lattice, with irreducibles graph JG($\mathbf{L}$).

Proposition 6.3. *There is a bijection between intervals* $(x, y) \in \mathbf{Int}_{\mathbf{L}}$ *and triples* (X, m, M) *where* X *is a min-max set of* G*,* $m \subseteq \min(X)$*,* $M \subseteq \max(X)$*, and* $m \sqcup M = X$*, which is given by :* (x, y) *has canonical join-representation*

$$\bigvee_{j \in m} (0, j) \vee \bigvee_{j \in M} (j, j).$$

Proposition 6.4. *There is a bijection between transvals* $(x, y) \in \mathbf{Trv}_{\mathbf{L}}$ *and triples* (X, m, M) *where* X *is a min-max set of* G*,* $m \subseteq \min(X)$*,* $M \subseteq \max(X)$*, and* $m \cup M = X$*, which is given by :* (x, y) *has canonical join-representation*

$$\bigvee_{j \in m} (0, j) \vee \bigvee_{j \in M} (j, j_*).$$

Intervals of $\mathbf{L}$ *correspond to triples* (X, m, M) *such that* $m = \min(X)$.

Proposition 6.5. *There is a bijection between faces $(x, y) \in \mathbf{FInt_L}$ and triples (X, m, M) where X is an independent set of G and $m \sqcup M = X$, which is given by : (x, y) has canonical join-representation*

$$\bigvee_{j \in m} (j_*, j) \vee \bigvee_{j \in M} (j, j).$$

Proposition 6.6. *There is a bijection between face transvals $(x, y) \in \mathbf{FTrv_L}$ and triples (X, m, M) where X is an independent set of G, and $m \cup M = X$, which is given by : (x, y) has canonical join-representation*

$$\bigvee_{j \in m} (j_*, j) \vee \bigvee_{j \in M} (j, j_*).$$

Faces of $\mathbf{L}$ *correspond to triples (X, m, M) such that $m = X$.*

6.1. **Enumeration of interval-closed subsets.**

Interval-closed sets = Convex sets. Maybe keep the name "interval-closed" to avoid confusion with convex topologies

— L.

If G is a two-acyclic factorization system which is transitive, it can be seen as a poset, and the corresponding lattice is distributive. The min-max sets of G are then in bijection with interval-closed subsets of G seen as a poset. In [ELM+24], the authors generalized rowmotion to interval-closed subsets of posets. It would be interesting to generalize this further on min-max subsets of semidistributive lattices. They showed that interval-closed subsets are in bijection with pairs of disjoint antichains (A, B) such that $B \subseteq\downarrow A$.

We define two different orders on the set of interval-closed subsets of a poset $\mathcal{P}$, and we prove that these orders are join-distributive lattices. Join-lattices can be represented by colored posets, generalizing the Birkhoff's representation theorem [HN18].

The reverse inclusion of interval-closed subsets yields a join-distributive lattice whose corresponding colored poset is

$$(\mathcal{P} \setminus \max(\mathcal{P})) \sqcup \overline{(\mathcal{P} \setminus \min(\mathcal{P}))},$$

where elements of $\mathcal{P}$ which are neither maximal nor minimal give two distinct elements with the same color.

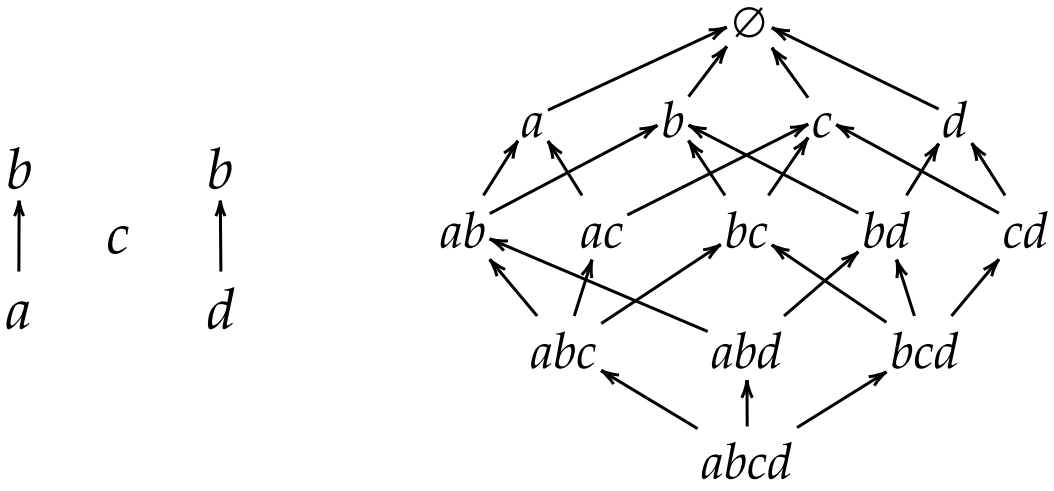


FIGURE 3

The order $(A_1, B_1) \leq (A_2, B_2) \iff \downarrow A_1 \subseteq \downarrow A_2$ and $\downarrow B_1 \subseteq \downarrow B_2$ yields a join-distributive lattice whose corresponding colored poset contains elements and edges of $\mathcal{P}$, with cover relations given by

$$x \lessdot y \lessdot z \in \mathcal{P} \implies \begin{cases} x \lessdot y \\ y \lessdot (x, y) \\ (x, y) \lessdot (y, z) \end{cases} .$$

Two edges have the same color if they have the same lower bound.

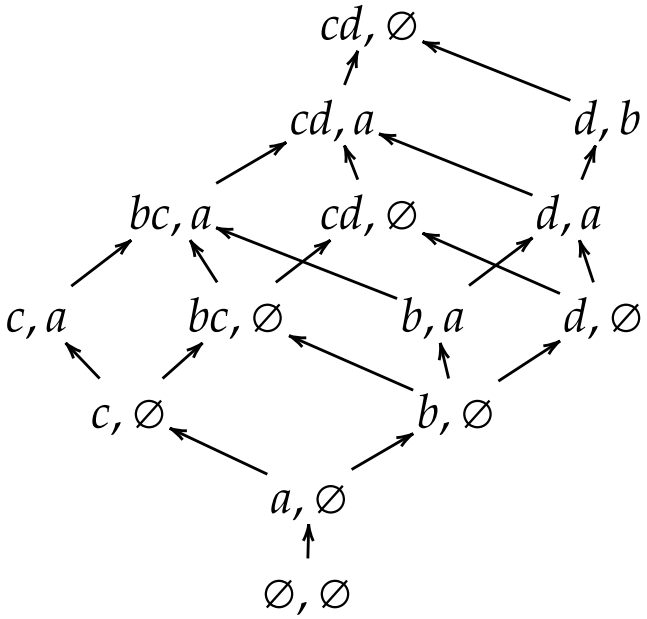


FIGURE 4

7. Transvals of the Tamari lattice

We show that transvals of the Tamari lattice $\mathbf{Tam}_n$ are in bijection with topologies on $\{1, ..., n\}$ having a base of convex sets. These topologies were first enumerated in [CR15].

Objects equienumerated or in bijection with convex topologies have already been studied in [BGP+26], [papier Jonathan Yang ?]

7.1. **Convex topologies.** Let X be a convex topology on $\{1, ..., n\}$. Each $k \in \{1, ..., n\}$ has a minimal neighborhood in X written $\mathrm{MN}_X(k)$ which is convex by definition. The mapping $k \mapsto \mathrm{MN}_X(k)$ characterizes the topology X.

We write $\mathbf{T_{cvx}}(n)$ the set of convex topologies on $\{1, \ldots, n\}$, with the following order: for all $X, Y \in \mathbf{T_{cvx}}(n)$, $X \leq Y$ if and only if for all $k \in \{1, ..., n\}$, $\mathrm{MN}_X(k) \leq \mathrm{MN}_Y(k)$ where minimal neighborhoods are compared coordinatewise. Nested convex topologies form a sublattice of convex topologies. The spine of ... has size $A001333(n)$

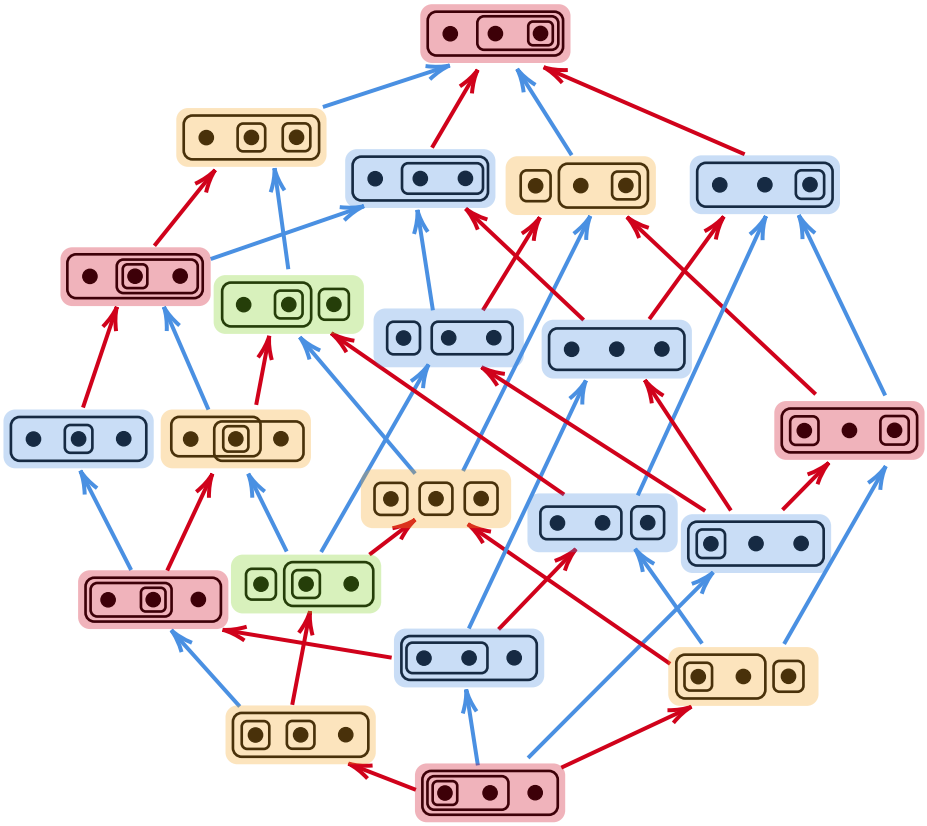


FIGURE 5. The lattice of convex topologies on $\{1, \dots, 3\}$.

We write $\mathbf{Tam}_n$ the Tamari lattice on binary trees of size n, *i.e.*, with n internal nodes. From a convex topology we construct recursively a binary tree in two different ways.

Definition 7.1. *Let $X \in \mathbf{T}_{\mathbf{cvx}}(n)$. If X is empty, $\mathrm{T}_{\mathrm{up}}(X)$ is the empty binary tree (a leaf). If not, $\mathrm{T}_{\mathrm{up}}(X)$ is the binary tree of size n whose left and right children are respectively $\mathrm{T}_{\mathrm{up}}(X \cap \{1, \dots, k-2\})$ and $\mathrm{T}_{\mathrm{up}}(X \cap \{k, \dots, n\})$ where $2 \leq k \leq n$ is the smallest value such that $\{k, \dots, n\}$ is open in X.*

If X is empty, $\mathrm{T}_{\mathrm{down}}(X)$ is the empty binary tree. If not, $\mathrm{T}_{\mathrm{down}}(X)$ is the binary tree of size n whose left and right children are respectively $\mathrm{T}_{\mathrm{down}}(X \cap \{1, \dots, k\})$ and $\mathrm{T}_{\mathrm{down}}(X \cap \{k+2, \dots, n\})$ where $1 \leq k \leq 1$ is the biggest value such that $\{1, \dots, k\}$ is open in X.

Proposition 7.2. *The map $X \mapsto (\mathrm{T}_{\mathrm{down}}(X), \mathrm{T}_{\mathrm{up}}(X))$ is an isomorphism from $\mathbf{T}_{\mathbf{cvx}}(n)$ to $\mathbf{Trv}_{\mathbf{Tam}_n}$.*

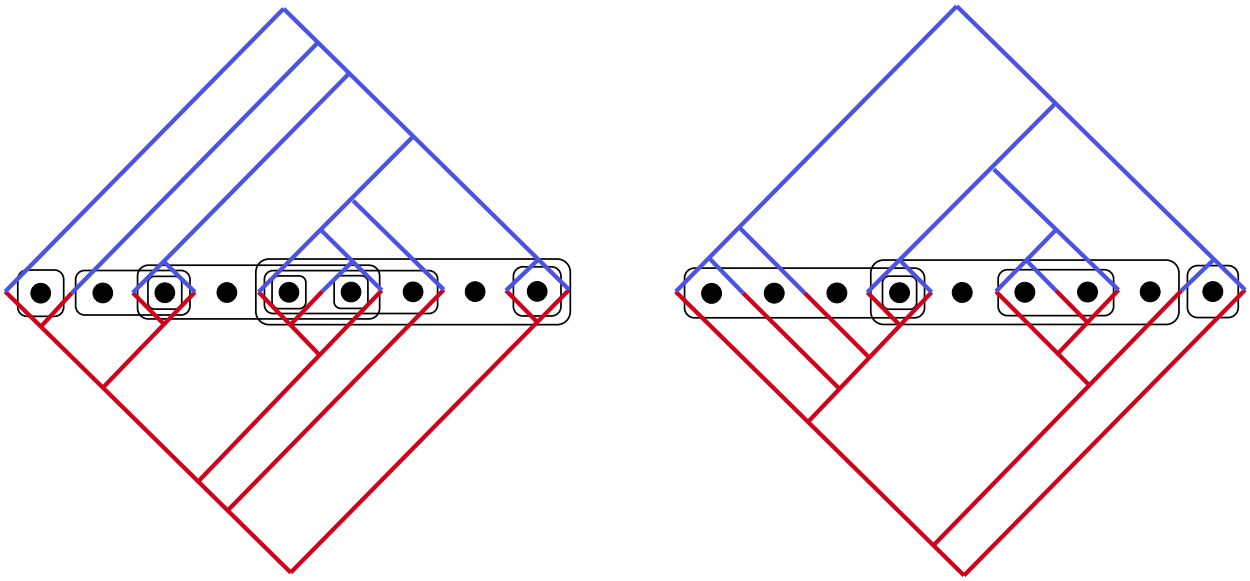


FIGURE 6. Two transvals of $\mathbf{Tam}_9$ with the corresponding convex topologies. The one on the left is an interval, the one on the right is not.

Proposition 7.3. *Let* $X \in \mathbf{T}_{\mathbf{cvx}}(n)$. $(\mathrm{T}_{\mathrm{down}}(X), \mathrm{T}_{\mathrm{up}}(X))$ *is an interval of* $\mathbf{Tam}_n$ *if and only the minimal neighborhoods* $\mathrm{MN}(k)$ *are all distinct for* $k \in \{1, \ldots, n\}$.

n	1	2	3	4	5	6	7	8	9	10
*A*000108	1	2	5	14	42	132	429	1430	4862	16796
*A*001003	1	3	11	45	197	903	4279	20793	103049	518859
*A*000260	1	3	13	68	399	2530	16965	118668	857956	6369883
*A*007564	1	4	19	100	562	3304	20071	124996	793774	5120632
*A*234268	1	4	21	129	876	6376	48829	388771	3191849	26864936

TABLE 2. The number of elements, face intervals, intervals, face transvals, and transvals of $\mathbf{Tam}_n$.

n	1	2	3	4	5	6	7	8	9	10
	1	0	1	0	2	0	5	0	14	0
	1	1	3	3	11	11	45	45	197	197
*A*369082	1	1	3	4	15	22	91	140	612	969
	1	2	5	10	28	56	167	334	1036	
	1	2	5	11	32	74	233	555	1833	

TABLE 3. The number of self-dual elements, face intervals, intervals, face transvals, and transvals of $\mathbf{Tam}_n$.

7.2. **Nested mountains.** MENTION TAMARI PREPOSETS (→ SECTION QUOTIENTS OF THE WEAK ORDER),

7.3. **Congruences of the lattice of transvals of $\mathbf{Tam}_n$.**

Proposition 7.4. *The lattice of congruences of* $\mathbf{Trv}(\mathbf{Tam}_n)$ *is isomorphic to the lattice of Motzkin paths of length* $2n-1$.

Proposition 7.5. $|\mathbf{FTrv}_{\mathbf{Tam}_n}| = |\mathbf{Cong}(\mathbf{FTrv}_{\mathbf{Tam}_n})|$

n	1	2	3	4	5	6	7	8	9	10
A000108	1	2	5	14	42	132	429	1430	4862	16796
A103211	1	4	28	244	2380	24868	272188	3080596	35758828	423373636
A001246	1	4	25	196	1681	2116	18769	168100	1510441	13586596
A007564	1	4	19	100	562	3304	20071	124996	793774	5120632
A099250	1	4	21	127	835	5798	41835	310572	2356779	18199284

TABLE 4. The number of congruences of the lattices of elements, face intervals, intervals, face transvals, and transvals of $\mathbf{Tam}_n$.

7.4. **The lattice of faces.** $\mathbf{Trv}(\mathbf{Tam}_n)$ contains the lattice of faces of $\mathbf{Tam}_n$ as a subposet of its intervals, FAUX (TAM4) but also as a sublattice containing nested convex topologies (cf. A007052)

7.5. **Transvals of Jurassian lattices.** Jurassian lattices do not have the same number of transvals or intervals as the Tamari lattice, but it is the case for those obtained from a rectangulation into chains.

8. Transvals of the weak order

Let $\mathfrak{g}$ be a finite-dimensional complex semisimple Lie algebra, W its Weyl group, and $U_q(\mathfrak{g})$ the quantized enveloping algebra of $\mathfrak{g}$. Heckenberger and Kolb [HK12] showed that homogeneous right coideal subalgebras of $U_q(\mathfrak{g})$ are in bijection with triples (x, u, J) where $x, u \in W$ with $u^{-1} \leq_R x$ and $J \subseteq \Pi \cap x\Pi$, where Π denotes a fixed basis of the root system of $\mathfrak{g}$. We show that they are also in bijection with transvals of the weak order.

Definition 8.1. *Let* $w \in W$. $(r, s) \in S^2$ *is a decoil of* w *if* $wr = sw$ *and* $l(sw) < l(w)$. *We write* $\mathrm{Dec}(w)$ *the set of decoils of* w.

Definition 8.2. *Let* $u, v \in W$ *such that* $u \leq_R v$. *We define* $\mathrm{AscDes}(u, v)$ *as the set containing all* $(r, s) \in S^2$ *such that* r *is an ascent of* u *and* s *a descent of* v *corresponding to the same reflection,* i.e. $u \leq_R ur$, $vs \leq v$ *and* $uru^{-1} = vsv^{-1}$.

Proposition 8.3. *Let* $u, v \in W$ *such that* $u \leq_R v$. *Elements of* $K_{[u,v]}$ *are in bijection with elements of* $\mathrm{AscDes}(u, v)$. *More precisely, we have* $(r, s) \in \mathrm{AscDes}(u, v)$ *if and only if* $ur \in K_{[u,v]}$, *and in this case* $\kappa(ur) = vs$.

Proposition 8.4. *Transvals of* $(W, \leq_R)$ *are in bijection with triples* (x, y, K) *where* $x \leq_R y$ *and* K *is a subset of* $\mathrm{Dec}(x)$.

Proof. By Propositions (3.9) and (8.3), transvals of $(W, \leq_R)$ are in bijection with triples (u, v, J) where $u \leq_R v$ and J is a subset of $\mathrm{AscDes}(u, v)$. We show that the map $(u, v) \mapsto (v^{-1}u, v^{-1})$ is a bijection between intervals of the right weak order, and that we have $\mathrm{AscDes}(u, v) = \mathrm{Dec}(v^{-1}u)$ for $u \leq_R v$. We have $u \leq_R v$ if and only u is a prefix of v, which is equivalent to $u^{-1}v$ being a suffix of v, *i.e.* $u^{-1}v \leq_L v$. By taking inverses, we get $v^{-1}u \leq_R v^{-1}$. Let $(r, s) \in \mathrm{AscDes}(u, v)$. Since s is a descent of v, it is also a descent of $u^{-1}v$ □

n	2	3	4	5	6	7	8	9
*A*000142	2	6	24	120	720	5040	40320	362880
*A*000670	3	13	75	541	4683	47293	545835	7087261
*A*007767	3	17	151	1899	31711	672697	17551323	549500451
*A*122704	4	22	160	1456	15905	202672	2951680	48361216
*A*213438	4	26	252	3368	58810	1290930	34604844	1107490596

TABLE 5. The number of elements, face intervals, intervals, face transvals, and transvals of $(A_n, \leq_R)$.

n	2	3	4	5	6	7
*A*000165	8	48	384	3840	46080	645120
*A*080253	17	147	1697	24483	423857	8560947
*A*395856	27	457	12069	449439	22145671	
	28	304	4432	80704	1763008	44932864
*A*395854	38	664	17848	672004	33369560	2094849020

TABLE 6. The number of elements, face intervals, intervals, face transvals, and transvals of $(B_n, \leq_R)$.

n	4	5	6	7
*A*002866	192	1920	23040	322560
*A*080254	865	12483	216113	4364979
*A*395857	3959	146649	7198141	447955715
	2320	42304	924352	23557888
*A*395855	6512	238720	11633624	720453984

TABLE 7. The number of elements, face intervals, intervals, face transvals, and transvals of $(D_n, \leq_R)$.

9. Transvals of quotients of the weak order

Bijection between transvals and preposets for any quotient, generalizing [CPP19]

Preposets on a finite set X are in bijection with topologies on X, by letting $x \blacktriangleleft y$ if and only if $\mathrm{MN}(x) \subseteq \mathrm{MN}(y)$, where $\mathrm{MN}(x)$ is the minimal neighborhood of x.

The forcing order of $\mathfrak{S}_n$ is isomorphic to the poset $\mathcal{F}_n$ defined as the set of arcs (a, A, B, b) with $1 \leq a < b \leq n$ and A, B a set partition of $\{a+1, \ldots, b-1\}$ (A corresponds to points below the arc), ordered by $(a_1, A_1, B_1, b_1) \leq (a_2, A_2, B_2, b_2)$ if $a_1 \leq b_1$, $A_1 \subseteq A_2$, $B_1 \subseteq B_2$ and $b_1 \geq b_2$.

Definition 9.1. *Let $I \in \mathbf{Low}(\mathcal{F}_n)$. A preposet $\blacktriangleleft$ is a weak order quotient preposet if and only if for all $a < b < c$ (or $a > b > c$), $a \blacktriangleleft c \implies a \blacktriangleleft b$ or $b \blacktriangleleft c$, and for all $(a, A, B, b) \in I$*

- *if $a \blacktriangleleft b$, there exists $k \in A$ such that $a \blacktriangleleft k$, or there exists $k \in B$ such that $k \blacktriangleleft b$;*
- *if $a \blacktriangleright b$, there exists $k \in A$ such that $k \blacktriangleright b$, or there exists $k \in B$ such that $a \blacktriangleright k$.*

The definition of weak order preposets translated on topologies, give the following:

If $a, c \in \mathrm{MN}(b)$ with $a < b < c$, then for all $a < k < b$, we have $k \in \mathrm{MN}(b)$ or $a \in \mathrm{MN}(k)$, and for all $b < k < c$ we have $k \in \mathrm{MN}(b)$ or $c \in \mathrm{MN}(k)$.

10. Acknowledgements

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